Agustina Díaz Peregalli

AI & Computer Vision R&D Engineer | Multimodal Perception & Edge AI

Paris, France | (On-site / Hybrid)

adiazperegalli@gmail.com | LinkedIn: Agustina Díaz | GitHub: Peregalli

PROFILE SUMMARY

Electrical Engineer and MVA Master's graduate (ENS Paris-Saclay) with strong foundations in signal processing, physical-AI, and multimodal machine learning. Brings 4+ years of industry R&D experience designing, training, and shipping perception models to production—spanning multimodal marine perception (RGB/Thermal, VideoMAE), physics-based parameter estimation, and real-time edge optimization (Qualcomm AI, TFLite) for compute- and power-constrained hardware. Motivated to build Pelagon's foundational multimodal perception stack for autonomous maritime defense.

TECHNICAL SKILLS

- **Multimodal Perception & ML:** Multimodal Sensor Fusion, Self-Supervised Learning, VideoMAE, Time-Series & Kinematic Modeling, 3D Vision, PyTorch, TensorFlow, OpenCV, MediaPipe
- **Edge AI & Low-Power Optimization:** Qualcomm AI Engine, TensorFlow Lite, Model Pruning & Quantization, MobileNet, Edge Inference, Docker, Linux
- **Signal Processing & Systems:** Signal Processing (Fourier, Filter Design, Low-SNR Analysis), State-Space Models (Kalman Filters), AWS (SageMaker, S3), ClearML, Git/GitHub
- **Programming Languages & Math:** Python, C++ (Basic), MATLAB, SQL, Convex Optimization, Probabilistic Graphical Models

WORK EXPERIENCE

Dassault Systèmes — Paris, France

Computer Vision R&D Engineer Intern — *May 2026 – Oct 2026*

- Researched and developed algorithms to estimate the internal volumetric filling of 3D Gaussian Splats (3D GS) from multi-view sensor captures.
- Modeled physical material properties and structural density to enable accurate **physics-based simulation** of reconstructed 3D assets.
- *Keywords: 3D Gaussian Splatting, Physical Parameter Estimation, Optimization, NanoBanana, VEO 3, Geometry Processing, Python.*

Digital Sense — Montevideo, Uruguay

R&D Engineer | Jr. R&D Engineer | R&D Assistant — *March 2021 – Sept 2025*

- **Multimodal Marine Perception (2025):** Architected and deployed an end-to-end multimodal perception system (**RGB & Thermal**) for maritime detection under adverse, low-SNR conditions. Built automated SageMaker/S3 pipelines and trained temporal self-supervised models using **VideoMAE**.
- **Edge AI & Hardware Optimization (2024–2025):** Developed an automated edge detection pipeline; trained custom models for adverse operational environments and optimized inference latency for compute-constrained edge hardware via Qualcomm AI & **Docker**.
- **Real-Time Edge CV & Production Delivery (2021–2024):** Engineered the core CV engine for a commercial application (Ulta App) serving high-volume retail users. Fine-tuned **MobileNet** models for landmarking/segmentation, compressed models for edge inference via TFLite pruning/quantization, and curated balanced datasets.
- *Keywords: Multimodal Fusion, Qualcomm AI, TensorFlow Lite, PyTorch, VideoMAE, AWS SageMaker, MobileNet, Docker, Edge Optimization.*

EDUCATION

École Normale Supérieure (ENS) — Paris, France

M.S. Mathematics, Vision and Learning (MVA) — *Sept 2025 – Sept 2026*

- **Coursework:** Convex Optimization, Probabilistic Graphical Models, 3D Computer Vision, Geometric Deep Learning.
- **Honors:** Awarded Uruguay's **ANII Postgraduate Studies Scholarship** for academic excellence to fund MVA studies.

Universidad de la República — Montevideo, Uruguay

B.S. Electrical Engineering — *March 2017 – March 2024*

- **Coursework:** Signal Processing, Stochastic Processes & Probability, Machine Learning & Pattern Recognition, Image Processing.

PUBLICATIONS & FEATURED PROJECTS

- **Publication (IBERAMIA 2024 / Springer 2025):** Zimmer, S., **Díaz, A.**, Aguilera, N., Estradé, M. J., Lecumberry, F., & Musé, P. *Evaluation of Mares Uterine Health Based on Endometrial Biopsies Using Image Processing and Machine Learning Techniques.* LNCS 15277, Springer. DOI: 10.1007/978-3-031-80366-6_13
- **Kalman VAE (MVA Research Project):** Reimplemented *"A Disentangled Recognition and Nonlinear Dynamics Model for Unsupervised Learning"* in PyTorch; analyzed latent disentanglement, variational inference bounds, and state-space models for kinematic sensor dynamics under noise.

LANGUAGES

Spanish: Native | **English:** Full Professional (B2+/C1) | **French:** Conversational | **Portuguese:** Intermediate